

Harshita Karnam

Engineering Undergraduate | Full-Stack Developer | Reinforcement Learning Researcher

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Portfolio: portfolio.vercel.app

PROFESSIONAL SUMMARY

Second-year B.Tech CSE student at IIIT Sri City with hands-on experience in full-stack web development and AI/ML research. Builds and deploys production-grade web applications using Next.js, React, Node.js, and MongoDB, and conducts RL research at the intersection of graph neural networks and large language models. Current manuscript SafeGAT, an LLM-augmented Graph Attention RL framework for adaptive traffic signal control, is under peer review. Known for strong problem-solving instincts, effective cross-functional collaboration, and a growth mindset that drives continuous learning. Communicates complex technical concepts clearly and thrives in fast-paced, team-oriented environments.

EDUCATION

B.Tech — Computer Science & Engineering 2024 – 2028
Indian Institute of Information Technology, Sri City, India

TECHNICAL SKILLS

Languages	Python, C, C++, HTML, CSS, JavaScript, TypeScript
Web / Full-Stack	React, Next.js, Node.js, REST APIs, JWT, MongoDB, Firebase, Vercel
ML / RL	PyTorch, OpenAI Gym, SUMO, DQN, DDQN, PPO, CoLight, Multi-Agent RL
Concepts	Graph Attention Networks, LLM Integration, Reward Shaping, Policy Gradient
Tools	Git / GitHub, Linux, VS Code

RESEARCH

SafeGAT (*Manuscript Under Review*) — LLM-augmented Graph Attention RL framework for adaptive traffic signal control; outperforms pure RL baselines on unseen road topologies.

PROJECTS

Velora Market — Multi-Vendor Marketplace *TypeScript, Next.js, MongoDB, JWT*

- Full-stack marketplace with buyer/seller/admin dashboards, role-based auth, and MongoDB persistence; Next.js API backend with HTTP-only refresh cookies.
- GitHub: [harshitak13/Velora](https://github.com/harshitak13/Velora) Live: velora-three-virid.vercel.app

CinePlex — React Movie Discovery App *React, Firebase, TMDB API*

- Trending film browser with genre search, trailers, personal watchlist, infinite scroll, and actor pages; Firebase Auth & Firestore for user data.
- GitHub: [harshitak13/cineplex](https://github.com/harshitak13/cineplex) Live: cineplex-aupw.vercel.app

SafeGAT — LLM-Integrated RL Traffic Signal Control *PyTorch, SUMO, LLM*

- Hybrid LLM + GAT-RL architecture for adaptive traffic control; improved generalisation across unseen intersections via LLM-derived auxiliary policy guidance.
- GitHub: [harshitak13/SafeGat](https://github.com/harshitak13/SafeGat)

RL Traffic Signal Control — DQN / DDQN / PPO / CoLight

- Benchmarked four RL algorithms on single and 4×4 multi-intersection grids; DDQN and CoLight led on reward, queue length, and travel time.
- GitHub: [harshitak13/RL-TSC](https://github.com/harshitak13/RL-TSC)

EXPERIENCE

Director, Management Department — Entrepreneurship Cell, IIIT Sri City *Sept 2025 – May 2026*
Member, Developer Relations — Google Developer Groups (GDG) *Sept 2025 – May 2026*